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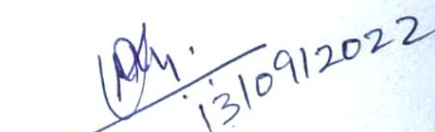
Lecture No.	Topics Covered	Total Page
9	Introduction of Vapour Power Cycles & Carnot Cycle	5
10	Working of Rankine cycle & Carnot Cycle	3
11	P-V & T-S Diagrams of Rankine Cycle & Carnot cycle	4
12	Comparison of Rankine cycle & Carnot Cycle on the basis of mean temp. of heat addition	7
13	Elementary Concept of Refrigeration Air Conditioning, Types of Refrigeration	5
14	Refrigeration Air Conditioning, Window air conditioning	5
15	Working principles and schematic diagrams of refrigerators	5
16	Air Conditioners and Ice Plants, Vapour compression cycle	4

References:

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons


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LECTURE -09(UNIT-02)

Syllabus :- Introduction of Vapour Power Cycles & Carnot Cycle

2.1 CARNOT CYCLE

Figure 2.1 shows a Carnot cycle on T-s and p-v diagrams. It consists of (i) two constant pressure operations (4-1) and (2-3) and (ii) two frictionless adiabatics (1-2) and (3-4). These operations are discussed below

1. **Operation (4-1).** 1 kg of boiling water at temperature T_1 is heated to form wet steam of dryness fraction x_1 . Thus heat is absorbed at constant temperature T_1 and pressure p_1 during this operation.

2. **Operation (1-2).** During this operation steam is expanded isentropically to temperature T_2 and pressure p_2 . The point '2' represents the condition of steam after expansion.

3. **Operation (2-3).** During this operation heat is rejected at constant pressure p_2 and temperature T_2 . As the steam is exhausted it becomes wetter and cooled from 2 to 3.

4. **Operation (3-4).** In this operation the wet steam at '3' is compressed isentropically till the steam regains its original state of temperature T_1 and pressure p_1 . Thus cycle is completed.

Refer T-s diagram :

Heat supplied at constant temperature T_1 [operation (4-1)] = area 4-1-b-a = $T_1 (s_1 - s_4)$ or $T_1 (s_2 - s_3)$.

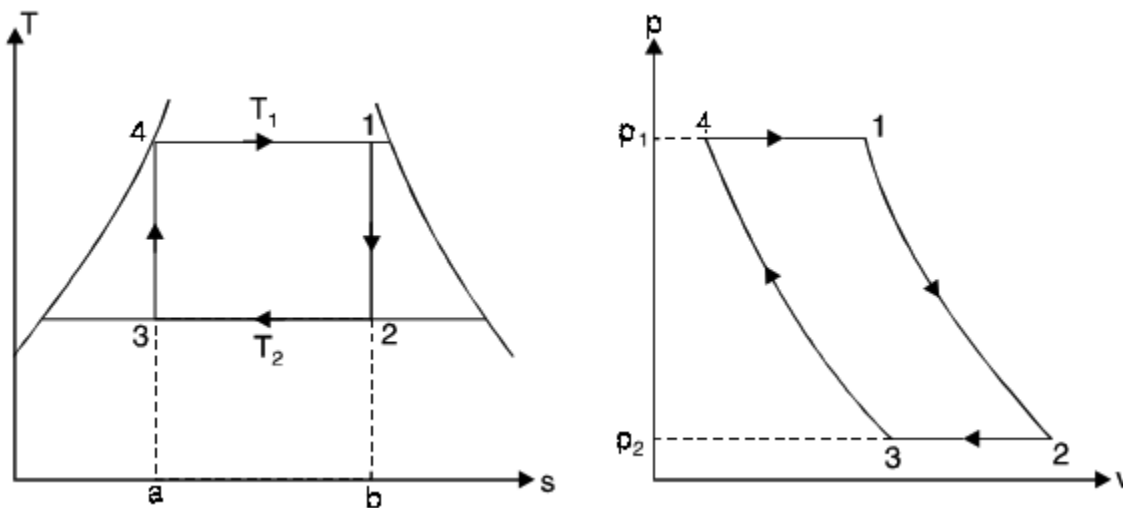


Fig 2.1 Carnot cycle on P-V & T-S diagram

Heat rejected at constant temperature T_2 (operation 2-3) = area 2-3-a-b = $T_2 (s_2 - s_3)$.

Since there is no exchange of heat during isentropic operations (1-2) and (3-4)

Net work done = Heat supplied – heat rejected

$$\begin{aligned} &= T_1 (s_2 - s_3) - T_2 (s_2 - s_3) \\ &= (T_1 - T_2) (s_2 - s_3). \end{aligned}$$

$$\begin{aligned} \text{Carnot cycle } \eta &= \frac{\text{Work done}}{\text{Heat supplied}} \\ &= \frac{(T_1 - T_2) (s_2 - s_3)}{T_1 (s_2 - s_3)} = \frac{T_1 - T_2}{T_1} \end{aligned} \quad \dots(12.1)$$

Limitations of Carnot Cycle

Though Carnot cycle is simple (thermodynamically) and has the highest thermal efficiency

for given values of T_1 and T_2 , yet it is extremely difficult to operate in practice because of the following reasons :

1. It is difficult to compress a wet vapour isentropically to the saturated state as required by the process 3-4.
2. It is difficult to control the quality of the condensate coming out of the condenser so that the state '3' is exactly obtained.
3. The efficiency of the Carnot cycle is greatly affected by the temperature T_1 at which heat is transferred to the working fluid. Since the critical temperature for steam is only 374°C , therefore, if the cycle is to be operated in the wet region, the maximum possible temperature is severely limited.
4. The cycle is still more difficult to operate in practice with superheated steam due to the necessity of supplying the superheat at constant temperature instead of constant pressure (as it is customary).

NOTE :-

In a practical cycle, limits of pressure and volume are far more easily realised than limits of temperature so that at present no practical engine operates on the Carnot cycle, although all modern cycles aspire to achieve it.

RANKINE CYCLE

Rankine cycle is the theoretical cycle on which the steam turbine (or engine) works.

The Rankine cycle is shown in Fig. 12.2. It comprises of the following processes :

Process 1-2 : Reversible adiabatic expansion in the turbine (or steam engine).

Process 2-3 : Constant-pressure transfer of heat in the condenser.

Process 3-4 : Reversible adiabatic pumping process in the feed pump.

Process 4-1 : Constant-pressure transfer of heat in the boiler.

Fig. 12.3 shows the Rankine cycle on p-v, T-s and h-s diagrams (when the saturated steam enters the turbine, the steam can be wet or superheated also).

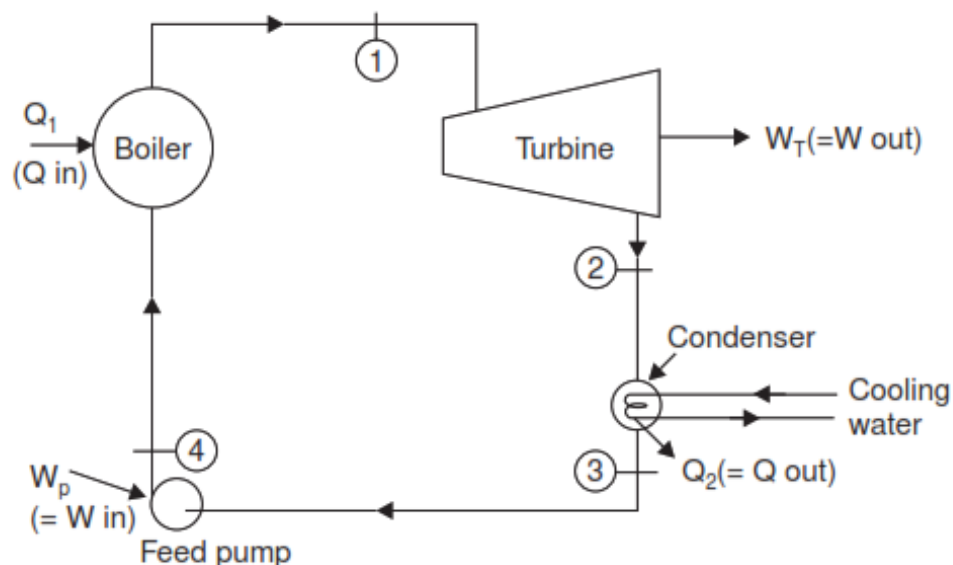


Fig. 12.2. Rankine cycle.

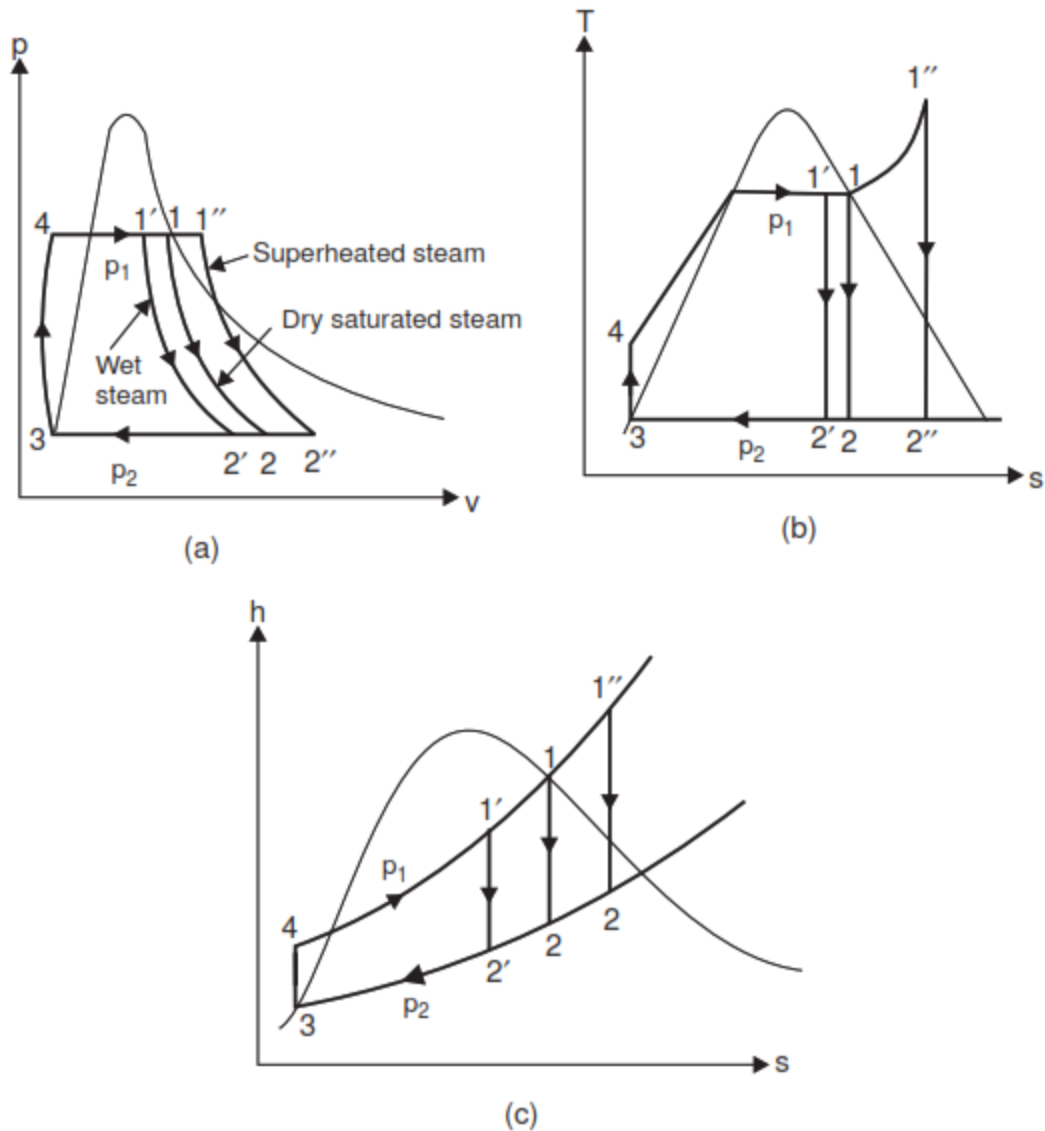


Fig. 12.3. (a) $p-v$ diagram ; (b) $T-s$ diagram ; (c) $h-s$ diagram for Rankine cycle.

Considering 1 kg of fluid :

Applying *steady flow energy equation* (S.F.E.E.) to boiler, turbine, condenser and pump :

(i) **For boiler** (as control volume), we get

$$h_{f_4} + Q_1 = h_1$$

$$\therefore Q_1 = h_1 - h_{f_4} \quad \dots(12.2)$$

(ii) **For turbine** (as control volume), we get

$$h_1 = W_T + h_2, \text{ where } W_T = \text{turbine work}$$

$$\therefore W_T = h_1 - h_2 \quad \dots(12.3)$$

(iii) **For condenser**, we get

$$h_2 = Q_2 + h_{f_3}$$

$$\therefore Q_2 = h_2 - h_{f_3} \quad \dots(12.4)$$

(iv) **For the feed pump**, we get

$$h_{f_3} + W_P = h_{f_4}, \quad \text{where, } W_P = \text{Pump work}$$

$$\therefore W_P = h_{f_4} - h_{f_3}$$

Now, efficiency of Rankine cycle is given by

$$\begin{aligned} \eta_{\text{Rankine}} &= \frac{W_{\text{net}}}{Q_1} = \frac{W_T - W_P}{Q_1} \\ &= \frac{(h_1 - h_2) - (h_{f_4} - h_{f_3})}{(h_1 - h_{f_4})} \quad \dots(12.5) \end{aligned}$$

The feed pump handles liquid water which is incompressible which means with the increase in pressure its density or specific volume undergoes a little change. Using general property relation for reversible adiabatic compression, we get

$$Tds = dh - vdp$$

$$\therefore ds = 0$$

$$\therefore dh = vdp$$

$$\text{or} \quad \Delta h = v \Delta p \quad \dots \text{ (since change in specific volume is negligible)}$$

$$\text{or} \quad h_{f_4} - h_{f_3} = v_3 (p_1 - p_2)$$

When p is in bar and v is in m^3/kg , we have

$$h_{f_4} - h_{f_3} = v_3 (p_1 - p_2) \times 10^5 \text{ J/kg}$$

The feed pump term $(h_{f_4} - h_{f_3})$ being a small quantity in comparison with turbine work, W_T , is usually neglected, *especially when the boiler pressures are low*.

$$\text{Then,} \quad \eta_{\text{Rankine}} = \frac{h_1 - h_2}{h_1 - h_{f_4}} \quad \dots[12.5 (a)]$$

REFERENCES:-

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons

LECTURE -10(UNIT-02)**Syllabus :-**

Carnot vapour power cycle, drawbacks as a reference cycle, simple Rankine cycle; description, T-s diagram, analysis for performance. Comparison of Carnot and Rankine cycles. Effects of pressure and temperature on Rankine cycle performance. Actual vapour power cycles. Ideal and practical regenerative Rankine cycles, open and closed feed water heaters. Reheat Rankine cycle.

Vapour power cycles are used in steam power plants. In a power cycle heat energy (released by the burning of fuel) is converted into work (shaft work), in which a working fluid repeatedly performs a succession of processes. In a vapour power cycle, the working fluid is water, which undergoes a change of phase.

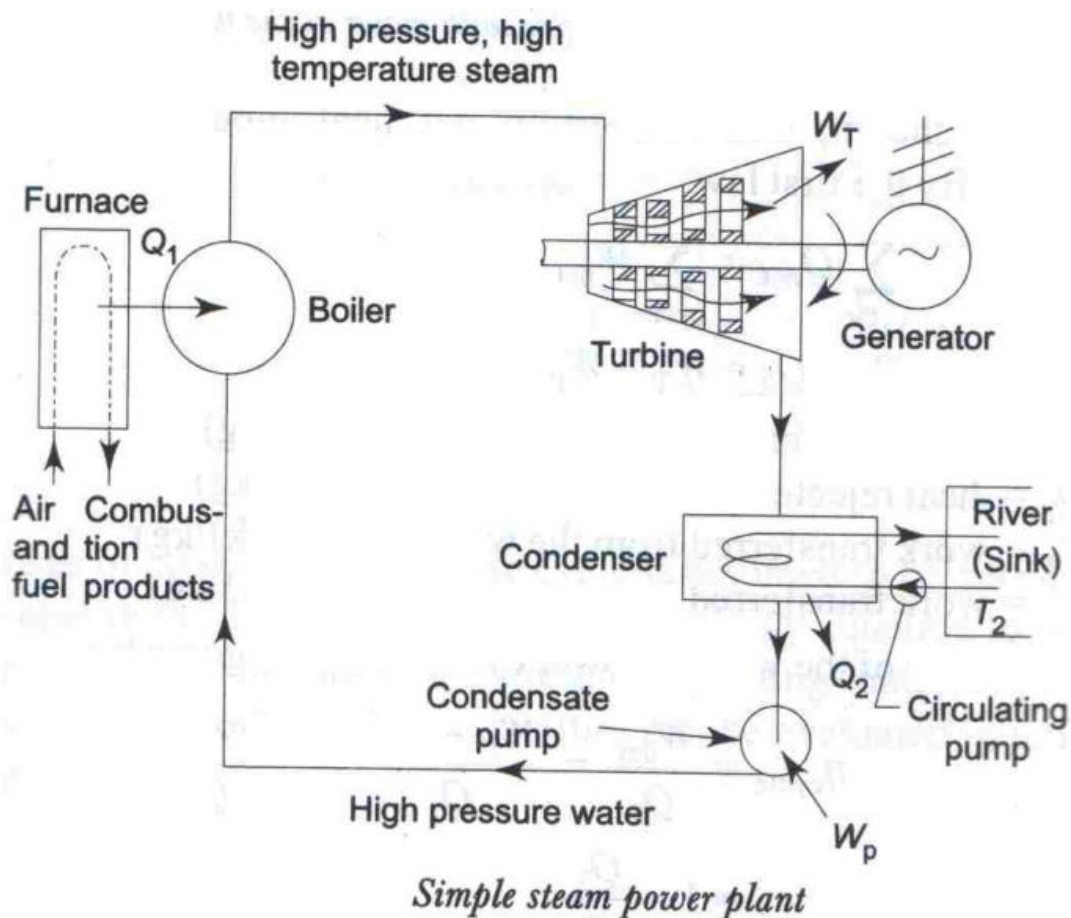


Figure shows a simple steam power plant working on the vapour power cycle. Heat is transferred to the water in the boiler (Q_H) from an external source. (Furnace, where fuel is continuously burnt) to raise steam, the high pressure high temperature steam leaving the boiler expands in the turbine to produce shaft work (W_T), the steam leaving the turbine condenses into water in the condenser (where cooling water circulates), rejecting heat (Q_L), and then the water is pumped back (W_P) to the boiler.

Since the fluid is undergoing a cyclic process, the net energy transferred as heat during the cycle must equal the net energy transfer as work from the fluid.

By the 1st law of Thermodynamics,

$$\sum_{cycle} Q_{net} = \sum_{cycle} W_{net}$$

Or $Q_H - Q_L = W_T - W_P$

Where Q_H = heat transferred to the working fluid (kJ/kg)

Q_L = heat rejected from the working fluid (kJ/kg)

W_T = work transferred from the working fluid (kJ/kg)

W_P = work transferred into the working fluid (kJ/kg)

$$\therefore \eta_{cycle} = \frac{W_{net}}{Q_H} = \frac{W_T - W_P}{Q_H} = \frac{Q_H - Q_L}{Q_H} = 1 - \frac{Q_L}{Q_H}$$

Idealized steam power cycles:

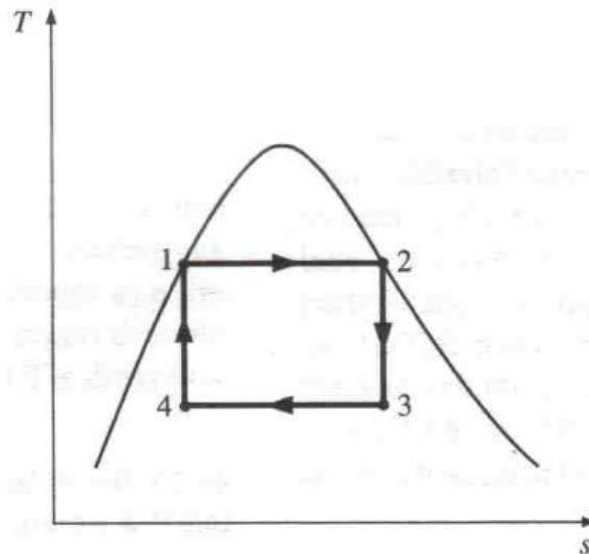


Fig:- T-S diagram of ideal Rankine cycle

We know that the efficiency of a Carnot engine is maximum and it does not depend on the working fluid. It is, therefore, natural to examine if a steam power plant can be operated on the Carnot cycle.

Figure shows the Carnot cycle on the T-S diagram. Heat addition at constant pressure P_2 , can be achieved isothermally in the process 1-2 in a boiler. The decrease in pressure from P_2 to P_3 in the process 2-3 can also be attained through the performance of work in a steam turbine. But in order to bring back the saturated liquid water to the boiler at the state 1, the condensation process 3-4 in the condenser must be terminated at the state 4, where the working fluid is a mixture of liquid water and vapour. But it is practically impossible to attain a condensation of this kind. Difficulty is also experienced in compressing isentropically the binary mixture from state 4 to the initial state 1, where the working fluid is entirely in the liquid state. Due to these inherent practical difficulties, Carnot cycle remains an ideal one.

REFERENCES:-

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons

LECTURE -11(UNIT-02)**Syllabus :-****Rankine cycle:-**

Rankine Cycle: The simplest way of overcoming the inherent practical difficulties of the Carnot cycle without deviating too much from it is to keep the processes 1-2 and 2-3 of the latter unchanged and to continue the process 3-4 in the condenser until all the vapour has been converted into liquid water. Water is then pumped into the boiler upto the pressure corresponding to the state 1 and the cycle is completed. Such a cycle is known as the Rankine cycle. This theoretical cycle is free of all the practical limitations of the Carnot cycle.

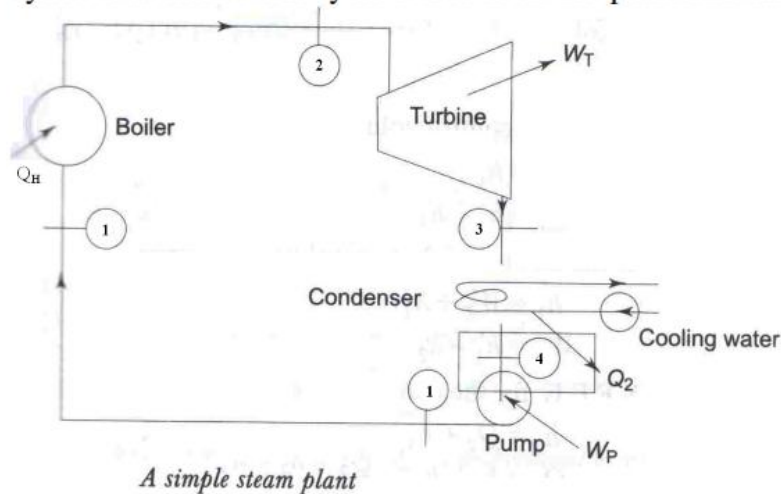


Figure (a) shows the schematic diagram for a simple steam power cycle which works on the principle of a Rankine cycle. Figure (b) represents the T-S diagram of the cycle.

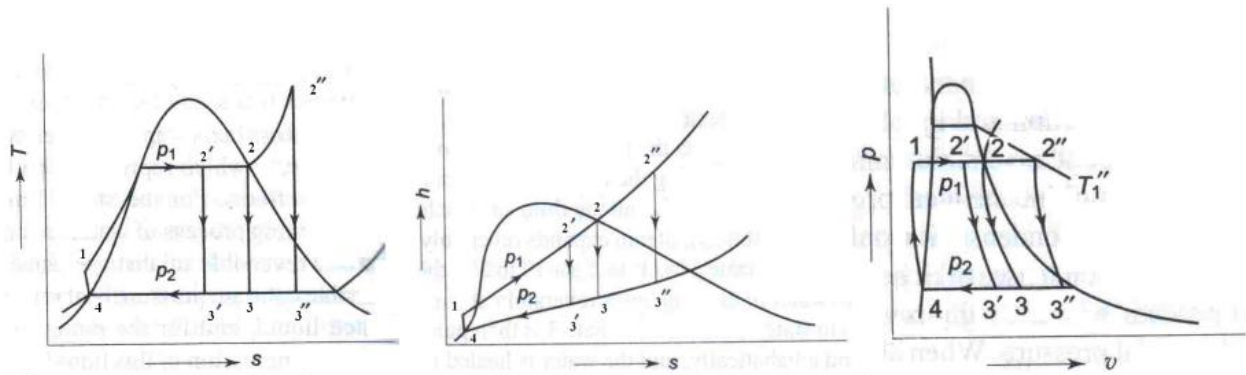
The Rankine cycle comprises the following processes.

Process 1-2: Constant pressure heat transfer process in the boiler

Process 2-3: Reversible adiabatic expansion process in the steam turbine

Process 3-4: Constant pressure heat transfer process in the condenser and

Process 4-1: Reversible adiabatic compression process in the pump.



The numbers on the plots correspond to the numbers on the schematic diagram. For any given pressure, the steam approaching the turbine may be dry saturated (state 2), wet (state 2') or superheated (state 2''), but the fluid approaching the pump is, in each case, saturated liquid (state 4). Steam expands reversibly and adiabatically in the turbine from state 2 to state 3 (or 2' to 3' or 2'' to 3''), the steam leaving the turbine condenses to water in the condenser reversibly at constant pressure from state 3 (or 3', or 3'') to state 4. Also, the water is heated in the boiler to form steam reversibly at constant pressure from state 1 to state 2 (or 2' or 2'')

Applying SFEE to each of the processes on the basis of unit mass of fluid and neglecting changes in KE & PE, the work and heat quantities can be evaluated.

For 1kg of fluid, the SFEE for the boiler as the CV, gives,

$$h_1 + Q_H = h_2 \quad \text{i.e., } Q_H = h_2 - h_1 \quad \text{--- (1)}$$

$$\text{SFEE to turbine, } h_2 = W_T + h_3 \quad \text{i.e., } W_T = h_2 - h_3 \quad \text{--- (2)}$$

$$\text{SFEE to condenser, } h_3 + Q_L = h_4 \quad \text{i.e., } Q_L = h_3 - h_4 \quad \text{--- (3)}$$

$$\text{SFEE to pump, } h_4 + W_P = h_1 \quad \text{i.e., } W_P = h_1 - h_4 \quad \text{--- (4)}$$

$$\text{The efficiency of Rankine cycle is } \eta = \frac{W_{net}}{Q_H} = \frac{W_T - W_P}{Q_H}$$

$$\text{i.e., } \eta = \frac{(h_2 - h_3) - (h_1 - h_4)}{(h_2 - h_1)} \quad \text{or} \quad \eta = \frac{(h_2 - h_1) - (h_3 - h_4)}{(h_2 - h_1)}$$

The pump handles liquid water which is incompressible i.e., its density or specific volume undergoes little change with an increase in pressure.

For reversible adiabatic compression, we have $Tds = dh - vdp$; since $ds = 0$

We have, $dh = vdp$

Since change in specific volume is negligible, $\Delta h = v \Delta P$

$$\text{Or } (h_1 - h_4) = v_4 (P_2 - P_3)$$

Usually the pump work is quite small compared to the turbine work and is sometimes neglected. In that case, $h_1 = h_4$

$$\eta_{\text{rankine}} \cong \frac{(h_2 - h_3)}{(h_2 - h_1)} \cong \frac{(h_2 - h_3)}{(h_2 - h_4)}$$

The efficiency of the Rankine cycle is presented graphically in the T-S diagram

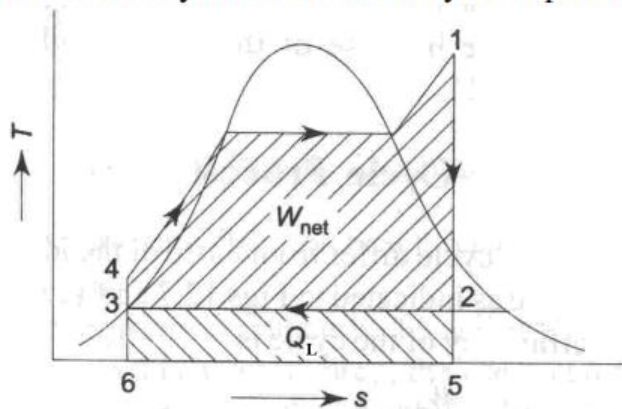


Fig. Q_H , W_{net} and Q_L are proportional to areas

$$Q_H \propto \text{area } 2-5-6-1,$$

$$Q_L \propto \text{area } 3-5-6-4$$

$$W_{\text{net}} = (Q_H - Q_L) = \text{area } 1-2-3-4 \text{ enclosed by the cycle.}$$

The capacity of the steam plant is expressed in terms of steam rate defined as the rate of steam flow (kg/h) required to produce unit shaft output (1kW)

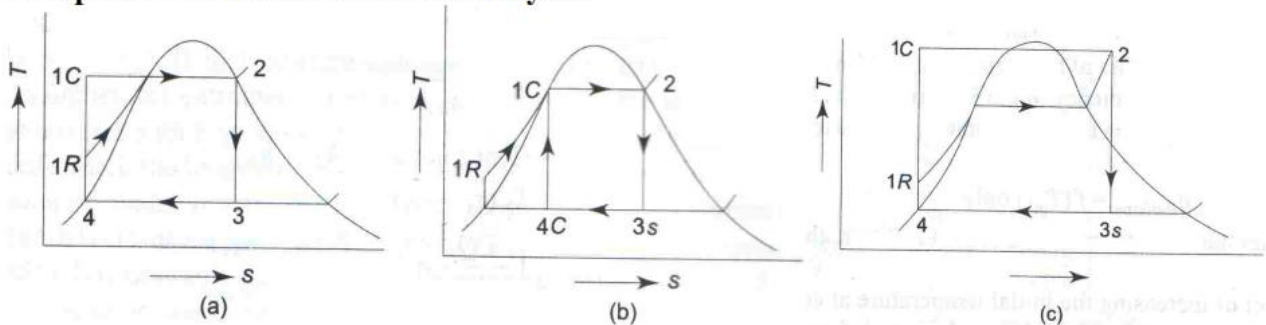
$$\begin{aligned} \therefore \text{Steam rate} &= \frac{1}{W_T - W_P} \frac{\text{kg}}{\text{kJ}} \frac{1\text{kJ/s}}{1\text{kW}} = \text{Specific Steam Consumption (SSC)} \\ &= \frac{1}{W_T - W_P} \frac{\text{kg}}{\text{kWs}} = \frac{3600}{W_T - W_P} \frac{\text{kg}}{\text{kWh}} \end{aligned}$$

The cycle efficiency also expressed alternatively as heat rate which is the rate of heat input (Q_H) required to produce unit work output (1kW)

$$\text{Heat rate} = \frac{3600 Q_H}{W_T - W_P} = \frac{3600}{\eta_{\text{cycle}}} \frac{\text{kJ}}{\text{kWh}}$$

$$\text{Lastly, work ratio } r_w = \frac{\oint \delta W}{\text{positive work}} = \frac{(h_2 - h_3) - (h_1 - h_4)}{(h_2 - h_3)}$$

Comparison of Rankine and Carnot cycles



Carnot cycle has the maximum possible efficiency for the given limits of temperature. But it is not suitable in steam power plants. Figure shows the Rankine and Carnot cycles on the T-S diagram.

The reversible adiabatic expansion in the turbine, the constant temperature heat rejection in the condenser, and the Reversible adiabatic compression in the pump, are similar characteristic features of both the Rankine and Carnot cycles. But whereas the heat addition process in the Rankine cycle is reversible and at constant pressure, in the Carnot cycle it is reversible and isothermal.

In Figures (a) and (c), Q_L is the same in both the cycles, but since Q_H is more, $\eta_C > \eta_R$. The two Carnot cycles in Figure (a) and (b) have the same thermal efficiency. $\therefore$ in Figure (b) also $\eta_C > \eta_R$.

But the Carnot cycle cannot be realized in practice because the pump work is very large. Whereas in (a) and (c) it is impossible to add heat at infinite pressures and at constant temperature from state 1C to state 2, in (b), it is difficult to control the quality at 4C, so that isentropic compression leads to a saturated liquid state.

REFERENCES:-

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons

LECTURE -12(UNIT-02)

Syllabus :-

Mean temperature of Heat addition

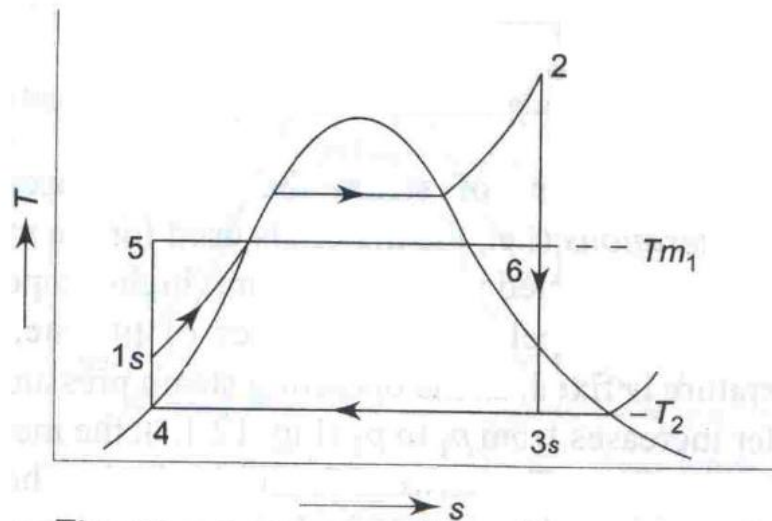


Fig. Mean temperature of heat addition

In the Rankine cycle, heat is added reversibly at a constant pressure, but at infinite temperature. Let T_{m1} is the mean temperature of heat addition, so that area under 1s and 2 is equal to the area under 5-6.

$$\text{Heat added, } Q_H = h_2 - h_{1s} = T_{m1} (S_2 - S_{1s})$$

$$\therefore T_{m1} = \text{Mean temperature of heat addition} = \frac{h_2 - h_{1s}}{S_2 - S_{1s}}$$

$$Q_L = \text{heat rejected} = h_{3s} - h_4 = T_3 (S_2 - S_{1s})$$

$$\therefore \eta_R = 1 - \frac{Q_L}{Q_H} = 1 - \frac{T_3 (S_2 - S_{1s})}{T_{m1} (S_2 - S_{1s})}$$

$$\eta_R = 1 - \frac{T_3}{T_{m1}} \quad \text{where } T_3 = \text{temperature of heat rejection.}$$

As T_3 is lowered for a given T_{m1} , the $\eta_R \uparrow$. But the lowest practical temperature of heat rejection is the ambient temperature T_0
 i.e., $\eta_R = f(T_{m1})$ only.

Or higher the mean temperature of heat addition, the higher will be the cycle efficiency.

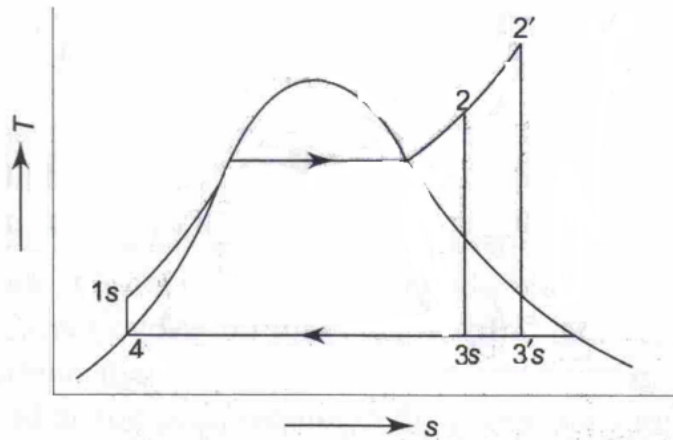


Fig. Effect of superheat on mean temperature of heat addition

The effect of increasing the initial temperature at constant pressure on cycle efficiency is shown in Figure. When the initial state changes from 2 to 2', T_{m1} between 2 and 2' is higher than T_{m1} between 1s and 2. So an increase in the superheat at constant pressure increases the mean temperature of heat addition and hence the cycle η .

But the maximum temperature of steam that can be used is fixed from metallurgical considerations (i.e., materials used for the manufacture of the components which are subjected to high pressure, high temperature steam such as super heaters, valves, pipelines, inlet stages of turbines etc).

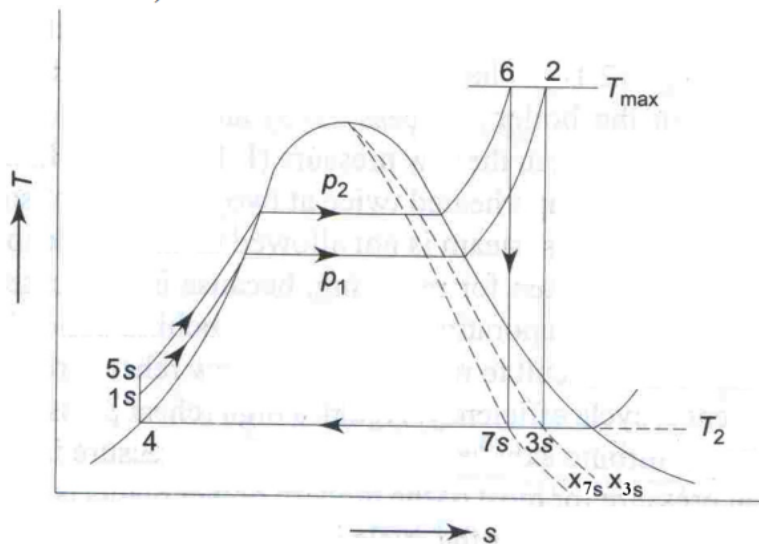


Fig. Effect of increase of pressure on Rankine cycle

When the maximum temperature is fixed, as the operating steam pressure at which heat is added in the boiler increases from P_1 to P_2 , the mean temperature of heat addition increases (since T_{m1} between 5s and 6 higher than between 1s and 2). But when the turbine inlet pressure increases from P_1 to P_2 , the ideal expansion line shifts to the left and the moisture content at the exhaust increases ($\because x_{7s} < x_{3s}$)

If the moisture content of steam in the turbine is higher the entrained water particles along with the vapour coming out from the nozzles with high velocity strike the blades and erode their surfaces, as a result of which the longevity of the blades decreases. From this consideration, moisture content at the turbine exhaust is not allowed to exceed 15% or $x < 0.85$.

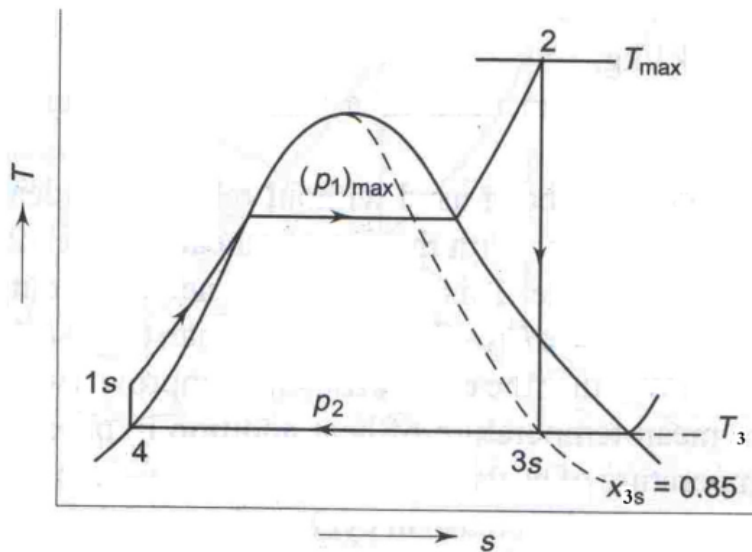


Fig. Fixing of exhaust quality, maximum temperature and maximum pressure in Rankine cycle

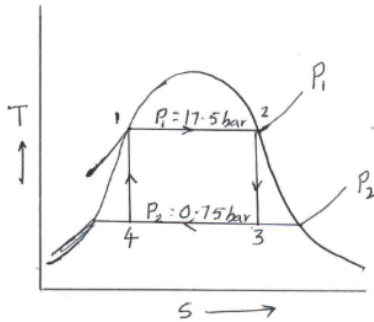
$\therefore$ With the maximum steam temperature at the turbine inlet, the minimum temperature of heat rejection and the minimum quality of steam at the turbine exhaust fixed, the maximum steam pressure at the turbine inlet also gets fixed. The vertical line drawn from 3S, fixed by T_3 and x_{3s} , intersects the T_{max} line, fixed by material, at 2, which gives maximum steam pressure at the turbine inlet.

Problems:

1. Dry saturated steam at 17.5 bar enters the turbine of a steam power plant and expands to the condenser pressure of 0.75 bar. Determine the Carnot and Rankine cycle efficiencies. Also find the work ratio of the Rankine cycle.

Solution: $P_1 = 17.5$ bar $P_2 = 0.75$ bar $\eta_{\text{Carnot}} = ?$ $\eta_{\text{Rankine}} = ?$

a) **Carnot cycle:** At pressure 17.5 bar from steam tables,



P	t_s	h_f	h_{fg}	h_g	S_f	S_{fg}	S_g
17	204.3	871.8	1921.6	2793.4	2.3712	4.0246	6.3958
18	207.11	884.5	1910.3	2794.8	2.3976	3.9776	6.3751

For $P = 17.5$ bar, using linear interpolation

$$\text{For } t_s, 204.3 + \frac{207.11 - 204.3}{1} \times 0.5 = 205.71^\circ \text{C}$$

$$= 478.71 \text{ K}$$

Similarly, $h_f = 878.15 \text{ kJ/kg}$ $h_{fg} = 1915.95 \text{ kJ/kg}$ $h_g = 2794.1 \text{ kJ/kg}$
 $S_f = 2.3844 \text{ kJ/kg}^\circ \text{K}$ $S_{fg} = 4.0011 \text{ kJ/kg}^\circ \text{K}$ $S_g = 6.3855 \text{ kJ/kg}^\circ \text{K}$

Also at pressure 0.75 bar from steam tables

P	t_s	h_f	h_{fg}	h_g	S_f	S_{fg}	S_g
0.8	93.51	391.7	2274.0	2665.8	1.233	6.2022	7.4352
0.7	89.96	376.8	2283.3	2660.1	1.1921	6.2883	7.4804

$\therefore$ For 0.75 bar, using linear interpolation,

$$t_s = 91.74^\circ \text{C} \quad h_f = 384.25 \quad h_{fg} = 2278.65 \quad h_g = 2662.95$$

$$S_f = 1.2126 \quad S_{fg} = 6.2453 \quad S_g = 7.4578$$

$$\text{The Carnot cycle } \eta, \eta_C = \frac{T_1 - T_2}{T_1} = \frac{478.71 - 364.74}{478.71} = 0.2381$$

$$\text{Steam rate or SSC} = \frac{1}{\oint \delta W} = \frac{1}{W_T - W_P}$$

Since the expansion work is isentropic, $S_2 = S_3$

But $S_2 = S_g = 6.3855$ and $S_3 = S_{f3} + x_3 S_{fg3}$

$$\text{i.e., } 6.3855 = 1.2126 + x_3 (6.2453) \quad \therefore x_3 = 0.828$$

$$\begin{aligned} \therefore \text{Enthalpy at state 3,} \quad h_3 &= h_{f3} + x_3 h_{fg3} \\ &= 384.25 + 0.828 (2278.65) = 2271.63 \text{ kJ/kg} \end{aligned}$$

$$\begin{aligned} \therefore \text{Turbine work or expansion work or positive work} &= h_2 - h_3 \\ &= 2794.1 - 2271.63 = 522.47 \text{ kJ/kg} \end{aligned}$$

Again since the compression process is isentropic

$$\text{i.e., } S_4 = S_1 = S_{f1} = 2.3844$$

$$\begin{aligned} \text{Hence } 2.3844 &= S_{f4} + x_4 S_{fg4} \\ &= 1.2126 + x_4 (6.2453) \quad \therefore x_4 = 0.188 \end{aligned}$$

$$\therefore \text{Enthalpy at state 4 is } h_4 = h_{f4} + x_4 h_{fg4}$$

$$\begin{aligned} &= 384.25 + 0.188 (2278.65) \\ &= 811.79 \text{ kJ/kg} \end{aligned}$$

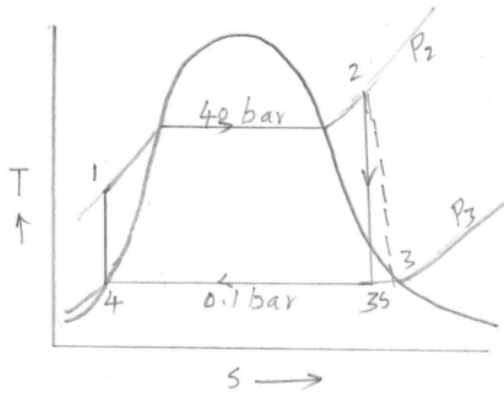
$$\begin{aligned} \therefore \text{Compression work, } &= h_1 - h_4 = 878.15 - 811.79 \\ W_P &= 66.36 \text{ kJ/kg} \end{aligned}$$

$$\therefore \text{SSC} = \frac{1}{522.47 - 66.36} = 2.192 \times 10^{-3} \text{ kg / kJ}$$

$$\text{work ratio} = r_w = \frac{\oint \delta w}{+ve \text{ work}} = \frac{W_T - W_P}{W_T} = \frac{456.11}{522.47} = 0.873$$

4. A 40 mW steam plant working on Rankine cycle operates between boiler pressure of 4 MPa and condenser pressure of 10 KPa. The steam leaves the boiler and enters the steam turbine at 400°C. The isentropic η of the steam turbine is 85%. Determine (i) the cycle η (ii) the quality of steam from the turbine and (iii) the steam flow rate in kg per hour. Consider pump work.

Solution:



$$P_2 = 4 \text{ MPa} = 40 \text{ bar} \quad P_3 = 10 \text{ KPa} = 0.1 \text{ bar}$$

$$P = 40000 \text{ kW} \quad t_2 = 400^\circ \text{C} \quad \eta_t = 0.85 \quad \eta_{\text{cycle}} = ? \quad x_3 = ?$$

$$\dot{m} = ?$$

$$h_2 = h|_{40 \text{ bar}, 400^\circ \text{C}} = 3215.7 \text{ kJ/kg} \text{ and } s_2 = 6.7733 \text{ kJ/kg-K}$$

$$h_4 = h_f|_{0.1 \text{ bar}} = 191.8 \text{ kJ/kg}$$

Process 2-3s is isentropic i.e., $S_2 = S_{3s}$

$$6.7733 = 0.6493 + x_{3s} (7.5018)$$

$$\therefore x_{3s} = 0.816$$

$$\therefore h_{3S} = h_{f3} + x_{3S} h_{fg3} = 191.8 + 0.816 (2392.9)$$

$$= 2145.2 \text{ kJ/kg}$$

$$\text{But } \eta_t = \frac{h_2 - h_3}{h_2 - h_{3S}} \quad \text{i.e., } 0.85 = \frac{3215.7 - h_3}{3215.7 - 2145.2}$$

$$\therefore h_3 = 2305.8 \text{ kJ/kg}$$

$$\therefore W_T = h_2 - h_3 = 3215.7 - 2305.8 = 909.9 \text{ kJ/kg}$$

$$W_P = v \int dp = 0.0010102 (40 - 0.1) 10^5 / 10^2$$

$$= 4.031 \text{ kJ/kg}$$

$$= h_1 - h_4 \quad \therefore h_1 = 195.8 \text{ kJ/kg}$$

$$(i) \eta_{cycle} = \frac{W_{net}}{Q_1} = \frac{909.9 - 4.031}{(3215.7 - 195.8)} = 29.9\%$$

$$(ii) x_3 = ? \quad \text{we have } 2305.8 = 191.8 + x_3 (2392.9) \quad \therefore x_3 = 0.88$$

$$(iii) P = \dot{m} W_{net} \quad \text{i.e., } 40000 = \dot{m} (905.87)$$

$$\therefore \dot{m} = 44.2 \text{ kg/s}$$

$$= 159120 \text{ kg/hr}$$

REFERENCES:-

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons

LECTURE -13(UNIT-2)**REFRIGERATION AND AIR CONDITIONING:****Syllabus:-**

Introduction, classification and types of refrigeration systems and air-conditioning. Applications of refrigeration and Air-conditioning.

INTRODUCTION

Refrigeration may be defined as the process of achieving and maintaining a temperature below that of the surroundings, the aim being to cool some product or space to the required temperature. One of the most important applications of refrigeration has been the preservation of perishable food products by storing them at low temperatures. Refrigeration systems are also used extensively for providing thermal comfort to human beings by means of air conditioning. Air Conditioning refers to the treatment of air so as to simultaneously control its temperature, moisture content, cleanliness, odour and circulation, as required by occupants, a process, or products in the space. The subject of refrigeration and air conditioning has evolved out of human need for food and comfort, and its history dates back to centuries. The history of refrigeration is very interesting since every aspect of it, the availability of refrigerants, the prime movers and the developments in compressors and the methods of refrigeration all are a part of it. The French scientist Roger Th venot has written an excellent book on the history of refrigeration throughout the world. Here we present only a brief history of the subject with special mention of the pioneers in the field and some important events.

UNIT OF REFRIGERATION

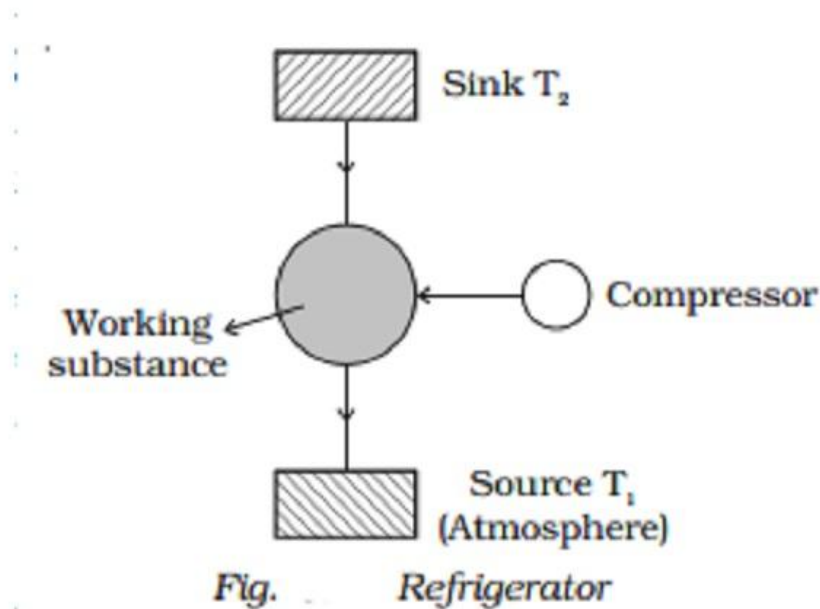
The **unit of refrigeration** is expressed in terms of ton of **refrigeration** (TR). One ton of **refrigeration** is defined as the amount of **refrigeration** effect (heat transfer rate) produced during uniform melting of one ton (100kg) of ice at 0 C to the water at the 0 C in 24 hours.

$$1\text{TR} = 3.5\text{KW} = 210 \text{ KJ/min}$$

COEFFICIENT OF PERFORMANCE

The coefficient of **performance** or COP (sometimes CP or CoP) of a heat pump, **refrigerator** or air conditioning system is a ratio of useful heating or cooling provided to work required. Higher COPs equate to lower operating costs.

Refrigerator : Coefficient of performance and Inferences

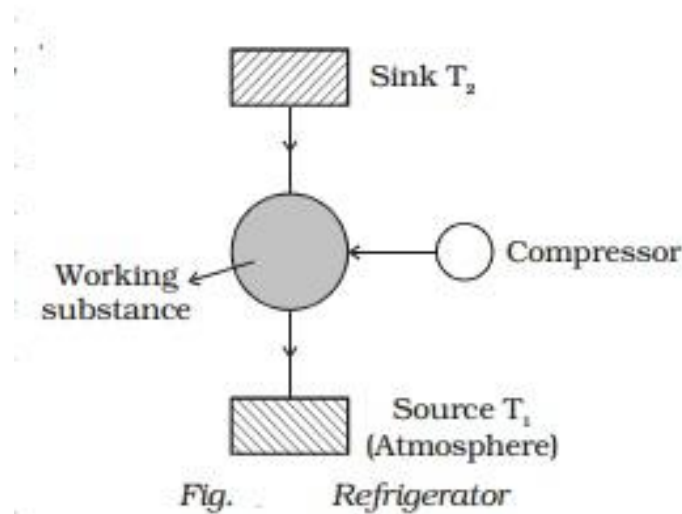


A refrigerator is a cooling device. An ideal refrigerator can be regarded as Carnot's heat engine working in the reverse direction.

Refrigerator

A refrigerator is a cooling device. An ideal refrigerator can be regarded as Carnot's heat engine working in the reverse direction. Therefore, it is also called a heat pump. In a refrigerator the working substance would absorb certain quantity of heat from the sink at lower temperature and reject a large amount of heat to the source at a higher temperature with the help of an external agency like an electric motor (Fig.).

In an actual refrigerator vapours of freon (dichloro difluoro methane CCl_2F_2) act as the working substance. Things kept inside the refrigerator act as a sink at a lower temperature T_2 . A certain amount of work W is performed by the compressor (operated by an electric motor) on the working substance. Therefore, it absorbs heat energy Q_2 from the sink and rejects Q_1 amount of heat energy to the source (atmosphere) at a temperature T_1 .



Since this is a reversible cyclic process, the change in the internal energy of the working substance is zero (i.e) $dU = 0$

According to the first law of thermodynamics,

$$dQ = dU + dW$$

$$\text{But } dQ = Q_2 - Q_1$$

$$dW = -W \text{ and } dU = 0$$

$$dQ = Q_2 - Q_1 = -W$$

Negative sign with W represents work is done on the system

$$\text{(i.e) } W = Q_1 - Q_2$$

Coefficient of performance

Coefficient of performance (COP) is defined as the ratio of quantity of heat Q_2 removed per cycle from the contents of the refrigerator to the energy spent per cycle W to remove this heat.

$$\text{(i.e) } COP = Q_2 / W$$

$$\text{COP} = Q_2 / (Q_1 - Q_2)$$

$$\text{(i.e) COP} = T_2 / (T_1 - T_2)$$

Application of refrigeration:

1. In water cooler to supply cold water
2. To manufacturing ice
3. For preservation of food, vegetables, milk, ice cream etc. in houses.
4. For preservation of chicken, mutton, fish etc.
5. For preservation of medicines, blood, tissues in hospital.
6. For preservation of dead bodies in hospital.
7. Different industrial application.

REFRIGERANT

A **refrigerant** is a substance or mixture, usually a [fluid](#), used in a [heat pump and refrigeration cycle](#). In most cycles it undergoes [phase transitions](#) from a [liquid](#) to a [gas](#) and back again. Many [working fluids](#) have been used for such purposes. [Fluorocarbons](#), especially [chlorofluorocarbons](#), became commonplace in the 20th century, but they are being phased out because of their [ozone depletion](#) effects. Other common refrigerants used in various applications are [ammonia](#), [sulfur dioxide](#), and non-halogenated [hydrocarbons](#) such as [propane](#)

The good properties of a refrigerant are as follows:

1. High critical temperature to have large isothermal energy transfer.
2. The specific heat of liquid should be as small as possible and of vapour should be as high as possible to give less superheating of vapour.
3. High latent heat of vaporization to get more refrigeration effect.
4. Large conductivity to reduce size of condenser and evaporator.
5. Low specific volume of vapour
6. Low freezing point such that there is no blockage during flow through evaporator.
7. Non corrosive to metal and inert so as to not react with other materials and commodities of the refrigeration system.
8. Non-toxic
9. Low boiling point

10. Non-flammable and non-explosive
11. Low cost and easily available
12. Easy to liquefy at moderate pressure and temperature
13. Easy of locating leaks by odour or suitable indicator
14. Mixes well with oil.
15. Low viscosity so that pressure drop is small.

TYPES OF REFRIGERANTS

The most common refrigerants used for air conditioning over the years include:

- **Chlorofluorocarbons (CFCs), including R12.** This is known to contribute to the greenhouse gas effect. Production of new stocks ceased in 1994.
- **Hydrochlorofluorocarbons (HCFCs), including R22.** Slightly less damaging to the ozone than R12, but the EPA has still mandated a phase out as a result of the Clean Air Act of 2010. R22 will phase out completely by 2020.
- **Hydrofluorocarbons (HFCs), including R410A and R134.** With no chlorine in the mix, this is safer for the environment and is now being used in place of R22. Air conditioners that run on R410A are more efficient, offer better air quality, increase comfort and improve reliability.

REFERENCES:-

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons

LECTURE -14(UNIT-2)**REFRIGERATION AND AIR CONDITIONING:****Syllabus:-****Air Conditioning System**

The Maintenance of controlled atmosphere according to the requirements in a space is known as “Air-conditioning”. The controlled atmosphere which gives maximum comfort to the human beings is known as comfort air-conditioning. The controlled atmosphere which is required for the manufacturing process is known as industrial air-conditioning. Comfort air-conditioning is further subdivided into summer air-conditioning and winter air-conditioning.

The air cooling and dehumidification used in summer is known as summer Air-conditioning and heating and humidification used in winter is known as winter air-conditioning.

Psychrometry

The Science which deals with the study of the behaviour of air and water vapour mixture is known as Psychrometry. The properties of water vapour and air mixture are known as

Psychrometric Properties.**Dry Air**

The pure dry air is a mixture of number of gases such as nitrogen, oxygen, and hydrogen and carbon dioxide. Density of air is 1.293 kg/m^3 at 1.01325 bar 273 K . Pure dry air does not exist in nature as it contains some impurities and some moisture all the time.

Moist Air

It is a mixture of dry air and water vapour. The amount of water vapour present in air depends on the pressure and temperature of the mixture.

Saturated Air

It is the mixture of dry air and water vapour when air has diffused maximum amount of watervapour into it. This means air has reached its moisture carrying capacity to the fullest and nomore addition is possible.

Specific Humidity (w)

It is the mass of water vapour present in 1kg of dry air and is expressed in terms of gram perkg of dry air.

Humidity Ratio

It is the same as specific humidity, i.e., mass of water vapour, present in unit mass of dry air.

Absolute Humidity

It is the mass of water vapour present in 1m³ of dry air and is expressed in terms of grams /m³of dry air. It can be expressed in terms of grains per cubic meter of dry air. 1kg of water vapour= 15,430 grains.

Relative Humidity

It is the ratio of actual mass of water vapour in a given volume of moist air to the mass of watervapour in the same volume of saturated air at the same temperature and pressure.

Specific Volume (v)

It is defined as the ratio of volume per unit mass of moist air.

Dry Bulb Temperature (DBT) or T_d

It is the temperature of air recorded by a thermometer when its bulb is surrounded by a wetcloth exposed to air.

Wet Bulb Depression (WBD)

It is the difference between dry bulb temperature and wet bulb temperature at any point.

Dew Point Temperature (DPT) or T_{dp}

It is the temperature of air recorded by a thermometer when water vapour in air begins to condense or dew is formed.

Classification of Air Conditioning Systems

The air conditioning may be broadly classified as follows:

1. According to the purpose

- Comfort air conditioning system, and
- Industrial air conditioning system,

2. According to the season of the year

- Winter air conditioning system,
- Summer air conditioning, and
- Year-round air conditioning.

3. According to the arrangement of equipment

- Unitary air conditioning system, and
- Central air conditioning system.

Window Air- Conditioner

Figure shows the schematic diagram of a window air-conditioner. The window air conditioner works on the vapour compression refrigeration cycle. The basic components of the system are compressor, condenser, a capillary tube and an evaporator, in addition, filters, accumulators, motors, fan and electrical controls from the rest of components. The high-pressure gas (refrigerant), which comes out of the compressor, is condensed in the condenser using ambient air as a coolant. The liquid refrigerant is expanded in a capillary tube and then enters the evaporator. Here, it evaporates and causes the refrigerating effect. Then fan blows the fresh air over the evaporator coil, which cools the air. The cool air is supplied to the room. Meanwhile, refrigerant from the evaporator enters the compressor to be compressed and the cycle repeats itself.

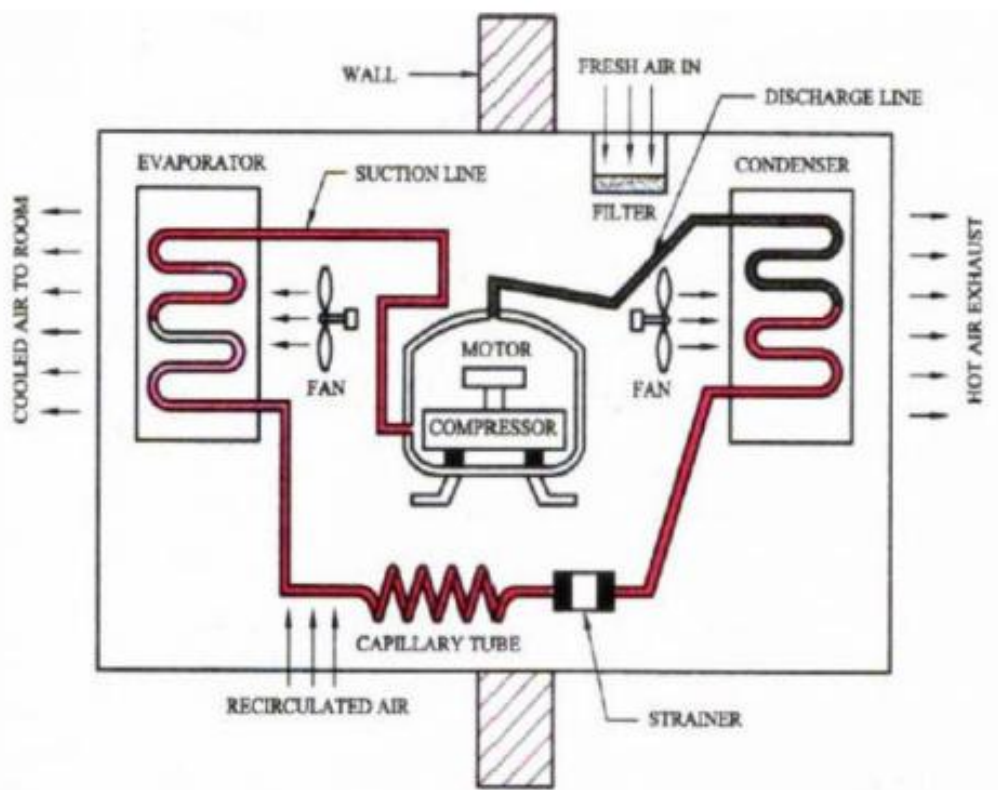


FIG. Window air conditioning

The return air from the room is exhausted by the evaporator fan by blowing air over the condensed coil, where it picks up heat from the refrigerant. The compressor used in a window air-conditioner is hermetic type, which is sealed in a casing with a motor as shown in the figure. The condenser and evaporator fans are both propeller type. The condenser coil is a continuous copper coil with aluminum fins. The capillary tube is 40cm long, usually with 0.75mm diameter. Evaporator coil is also made of copper with aluminum fins. Dampers and louvers are provided for intake and exhaust on both sides of the air-conditioner casing. A one tone window air-conditioner circulates $4\text{m}^3/\text{min}$ of air in the room. Some fraction of air (25%) is fresh; whereas some fraction (75%) of air is recirculated. Filters are used to filter air entering the A/C unit. The filter is plastic foam or metallic mat type.

REFERENCES: -

1. R. K. Rajput, Basic Mechanical Engineering, Laxmi Publications
2. B. Agrawal, Mechanical Engineering, John Wiley & Sons

LECTURE -15 (UNIT-2)**TYPES OF REFRIGERATION SYSTEMS AND AIR-CONDITIONING.****Split Air conditioning System****Split Type Room Air-Conditioner**

These are preferred for offices, schools, auditoriums, etc. The idea of split units came from the fact that air-conditioning system operation was noisy. In order to eliminate the noise, compressor and condenser were located at remote location and only evaporator coil and a fan were located near the application. This division or splitting of the air-conditioning unit gave the equipment its name. The given figure shows the schematic of a split unit. Usually split units are wall mounted. They are preferred due to their noiseless operation. The refrigerant liquid lines and suction lines are line gas they connect the compressor at the remote location to the split unit evaporator. This increases cost of the system. This operation is similar to the window air-conditioner as the basic cycle of operation is vapour compression refrigeration. The refrigerant is compressed, condensed, expanded at the remote site and evaporated only in the room evaporator. The air is blown over the evaporator coil and that causes cooling.

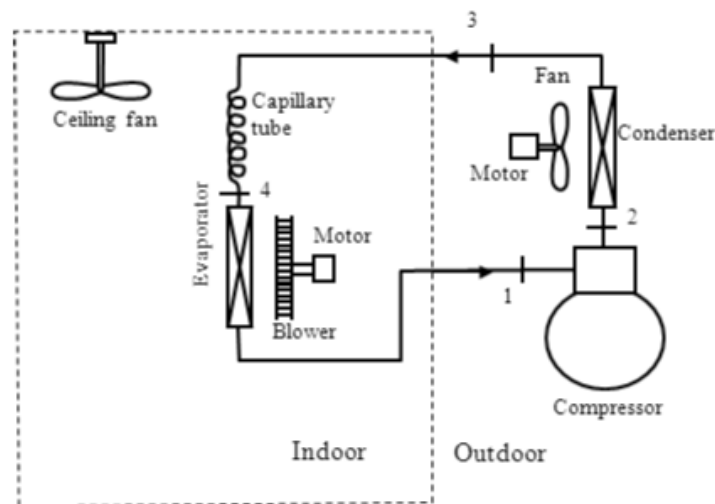


Fig. 1 Split A.C with Capillary tube inside the room

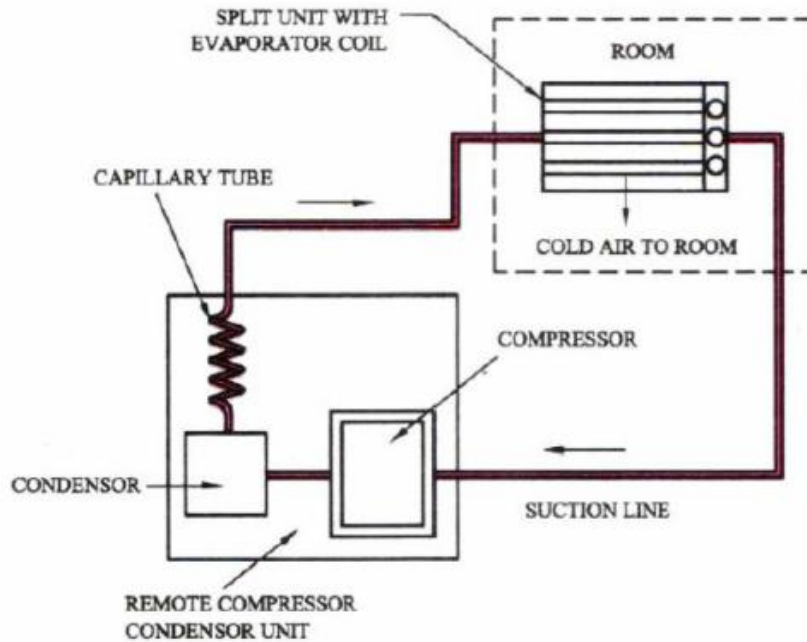


Fig. 2 Split A.C with Capillary tube outside the room

Advantages of the Units are

- _ Noiseless operation
- _ Easy Maintenance
- _ Less Vibration
- _ Compact unit
- _ Aesthetic quality of unit is high
- _ Disadvantages of the Units are
- _ High cost of unit
- _ Longer refrigerant lines and thus more leakage
- _ COP of unit is poor
- _ Control over operation is difficult

Applications of Refrigeration

- Food processing, preservation and distribution
- Chemical and process industries
- Comfort air-conditioning

SIMPLE VAPOR COMPRESSION REFRIGERATION SYSTEM

A simple vapor compression refrigeration system consists of the following equipment:

i) Compressor ii) Condenser iii) Expansion valve iv) Evaporator.

The vapour at low temperature and pressure enters the compressor where it is compressed isentropically and subsequently its temperature and pressure increase considerably. This vapour after leaving the compressor enters the condenser where it is condensed into high pressure liquid and is collected in a receiver tank. From receiver tank it passes through the expansion valve. Here it is throttled down to a lower pressure and has a low temperature. After finding its way through expansion valve it finally passes on to evaporator where it extracts heat from the surroundings or circulating fluid being refrigerated and vaporises to low pressure vapour.

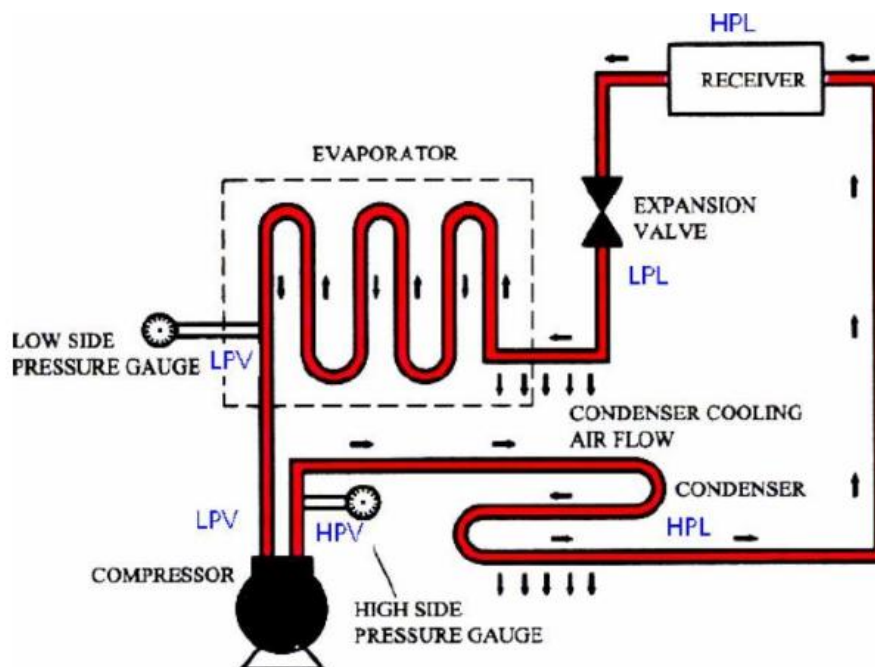


Fig. Vapour compression refrigeration system

State merits and demerits of an air refrigeration system.**Merits:**

- Air is easily available
- It is cheap
- Weight per TR is low and hence ideal for aircraft refrigeration
- No leakage problem
- It is non-flammable and non-toxic refrigerant
- Low maintenance cost

Demerits:

- It absorbs heat as sensible heat only, hence mass flow rate required is more as compared to other refrigerants
- Low COP
- Freezing of moisture at low temperature.

SIMPLE VAPOUR ABSORPTION SYSTEM

Some liquids like water have great affinity for absorbing large quantities of certain vapours (NH_3) and reduce the total volume greatly. The absorption refrigeration system differs fundamentally from vapor compression system only in the method of compressing the refrigerant. An absorber, generator and pump in the absorption refrigerating system replace the compressor of a vapor compression system. Figure shows the schematic diagram of a vapor absorption system. Ammonia vapor is produced in the generator at high pressure from the strong solution of NH_3 by an external heating source. The water vapor carried with ammonia is removed in the rectifier and only the dehydrated ammonia gas enters into the condenser.

High pressure NH_3 vapor is condensed in the condenser. The cooled NH_3 solution is passed through a throttle valve and the pressure and temperature of the

refrigerant are reduced below the temperature to be maintained in the evaporator. The low temperature refrigerant enters the evaporator and absorbs the required heat from the evaporator and leaves the evaporator as saturated vapor. Slightly superheated, low pressure NH_3 vapor is absorbed by the weak solution of NH_3 which is sprayed in the absorber as shown. Weak NH_3 solution (aqua-ammonia) entering the absorber becomes strong solution after absorbing NH_3 vapor and then it is pumped to the generator through the heat exchanger. The pump increases the pressure of the strong solution to generator pressure. The strong NH_3 solution coming from the absorber absorbs heat from high temperature weak NH_3 solution in the heat exchanger. The solution in the generator becomes weak as NH_3 vapor comes out of it.

The weak high temperature ammonia solution from the generator is passed to the heat exchanger through the throttle valve. The pressure of the liquid is reduced to the absorber pressure by the throttle valve.

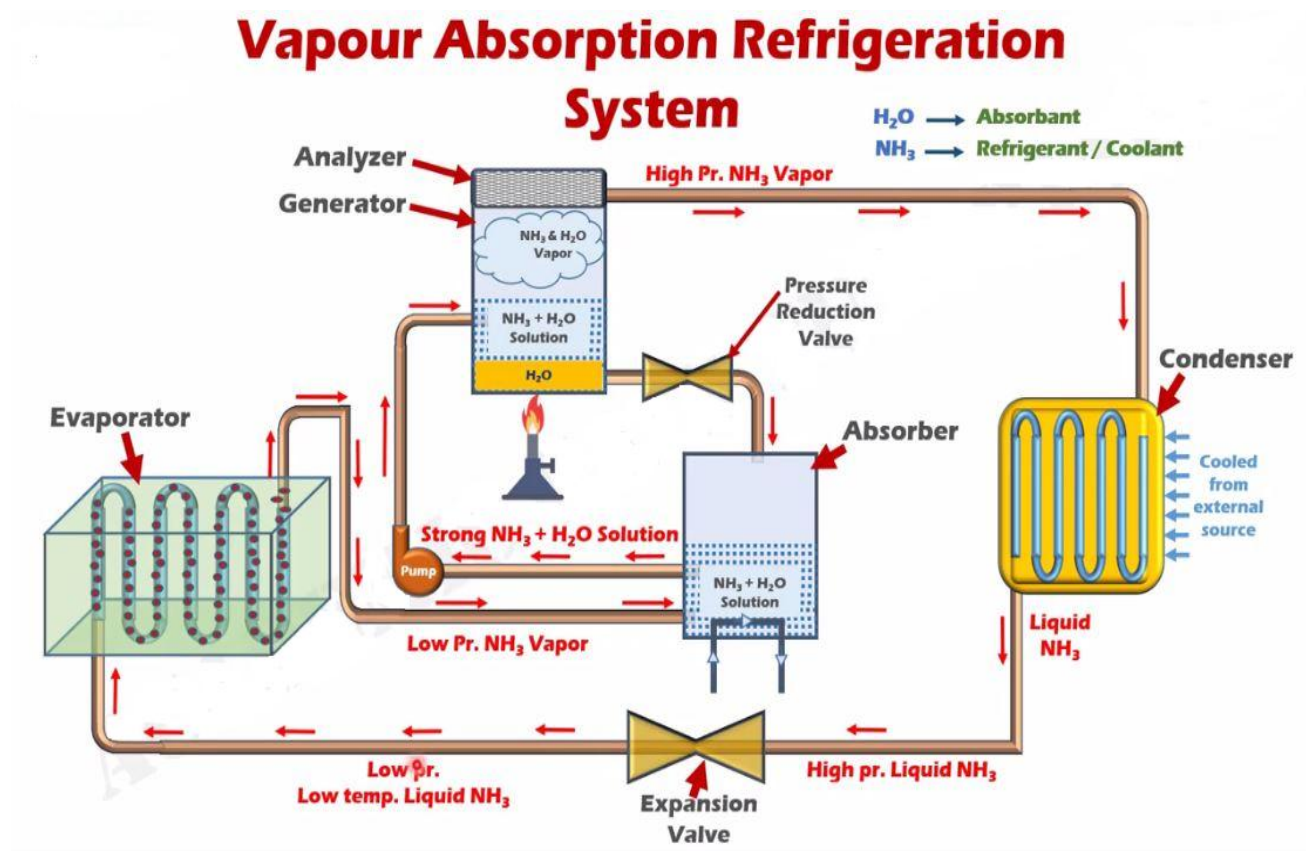


Fig. 3 Vapour Absorption RAC system

LECTURE -16(UNIT-2)**SIMPLE VAPOR COMPRESSION REFRIGERATION SYSTEM**

A simple vapor compression refrigeration system consists of the following equipment:

i) Compressor ii) Condenser iii) Expansion valve iv) Evaporator.

The vapour at low temperature and pressure enters the compressor where it is compressed isentropically and subsequently its temperature and pressure increase considerably. This vapour after leaving the compressor enters the condenser where it is condensed into high pressure liquid and is collected in a receiver tank. From receiver tank it passes through the expansion valve. Here it is throttled down to a lower pressure and has a low temperature. After finding its way through expansion valve it finally passes on to evaporator where it extracts heat from the surroundings or circulating fluid being refrigerated and vaporises to low pressure vapour.

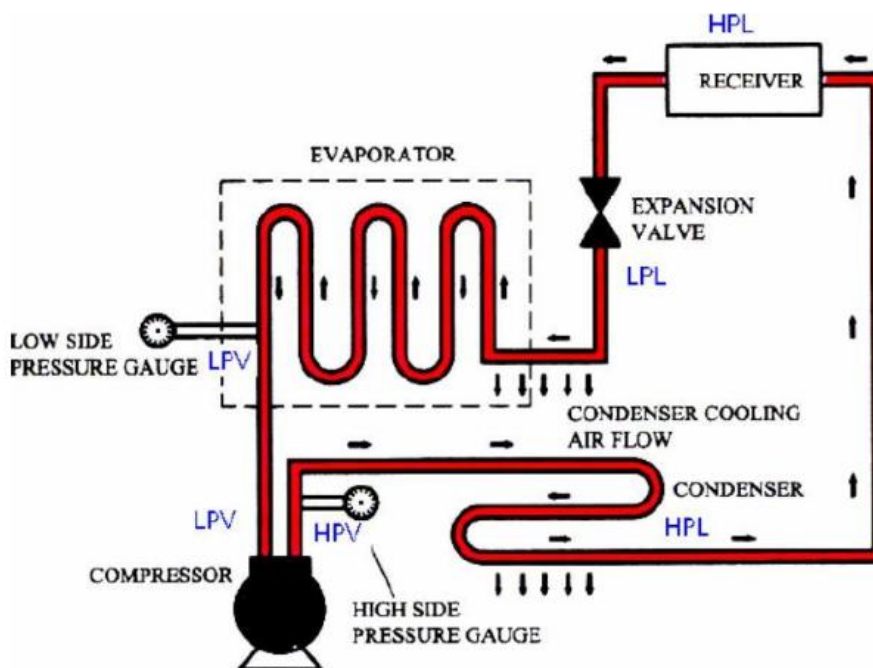


Fig. Vapour compression refrigeration system

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- Weight per TR is low and hence ideal for aircraft refrigeration
- No leakage problem
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Demerits:

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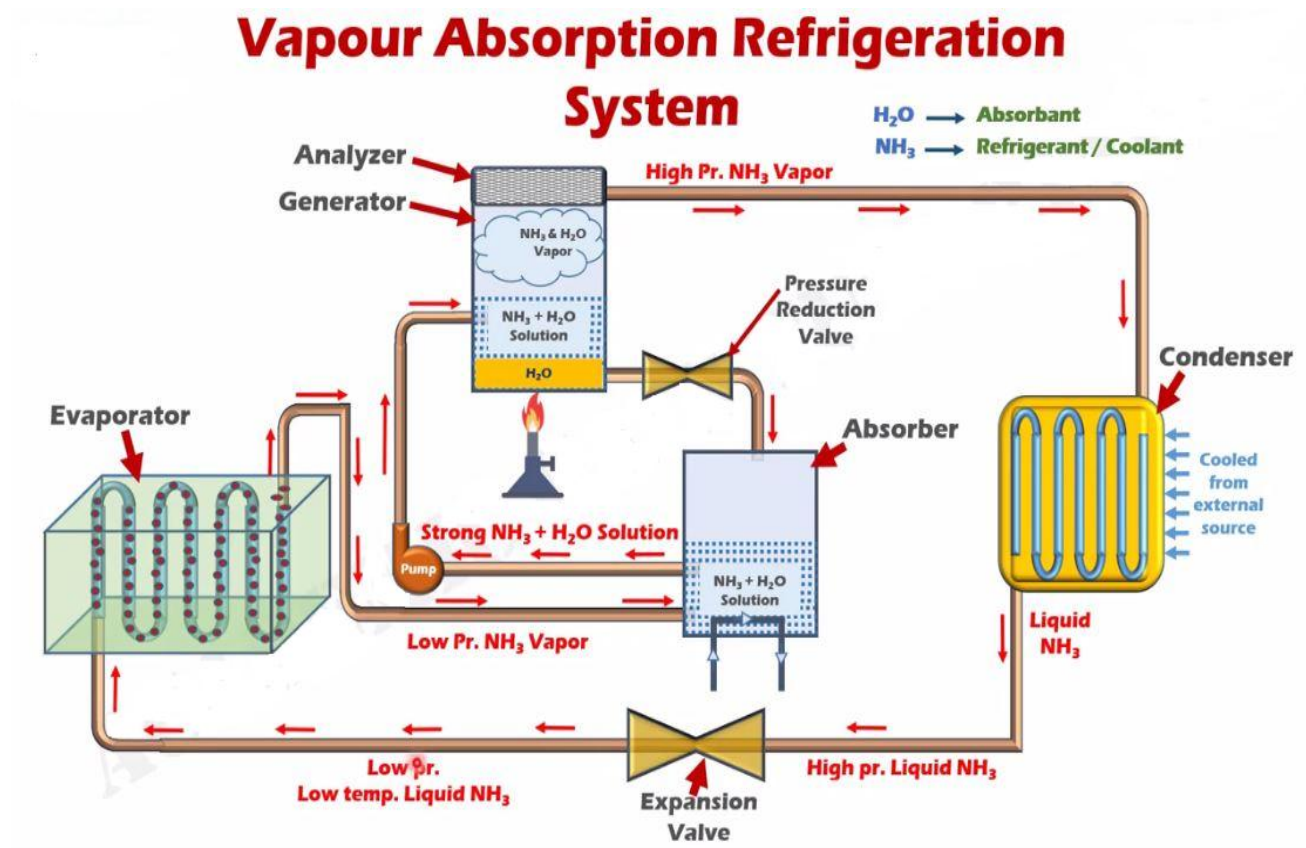


Fig. 3 Vapour Absorption RAC system